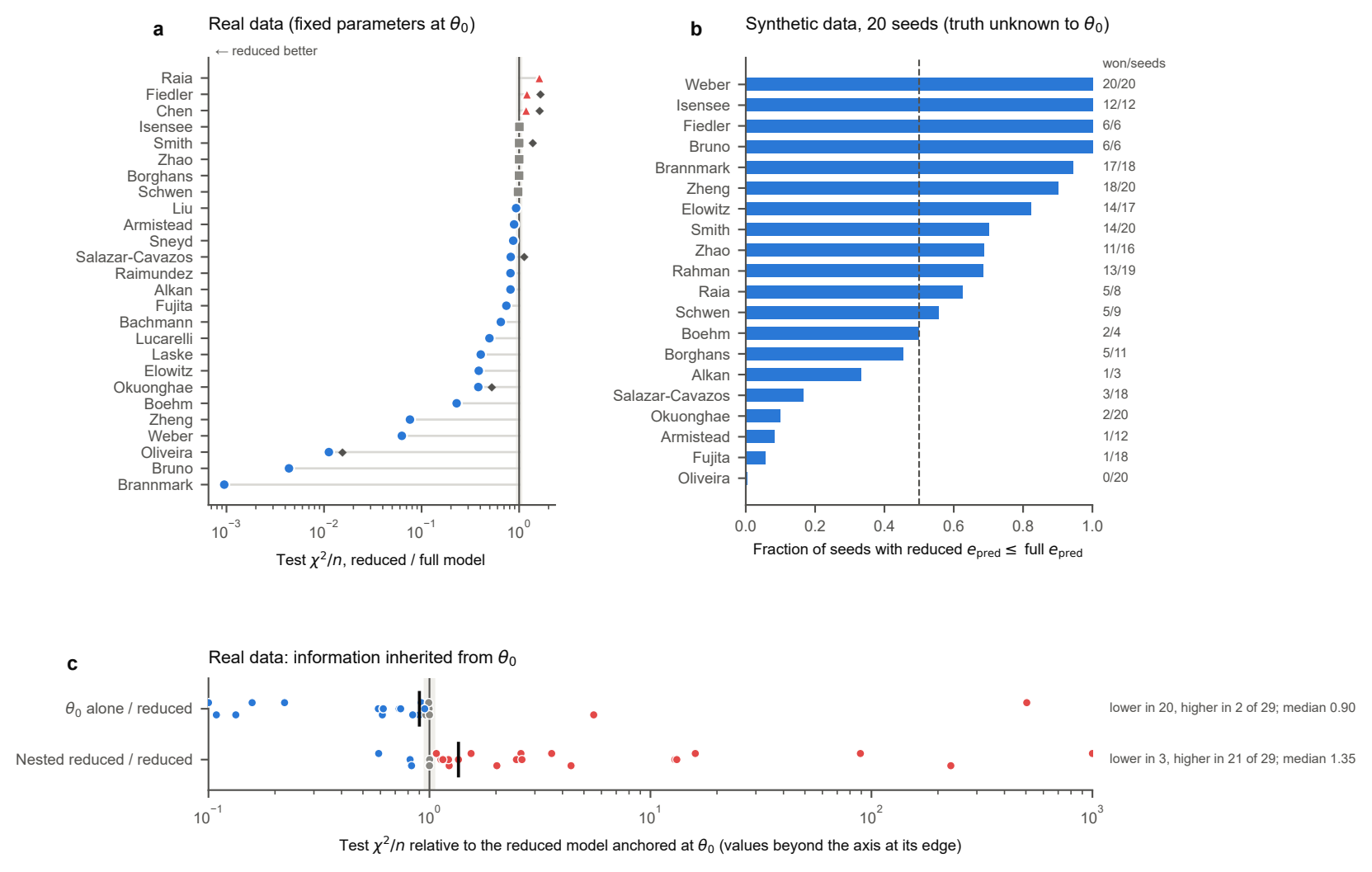

a
Real data (fixed parameters at θ_0)
← reduced better
Raia
Fiedler
Chen
Isensee
Smith
Zhao
Borghans
Schwen
Liu
Armistead
Sneyd
Salazar-Cavazos
Raimundez
Alkan
Fujita
Bachmann
Lucarelli
Laske
Elowitz
Okuonghae
Boehm
Zheng
Weber
Oliveira
Bruno
Brannmark
10^{-3}
10^{-2}
10^{-1}
10^{0}
Test χ^2/n, reduced / full model
b
Synthetic data, 20 seeds (truth unknown to θ_0)
won/seeds
Weber 20/20
Isensee 12/12
Fiedler 6/6
Bruno 6/6
Brannmark 17/18
Zheng 18/20
Elowitz 14/17
Smith 14/20
Zhao 11/16
Rahman 13/19
Raia 5/8
Schwen 5/9
Boehm 2/4
Borghans 5/11
Alkan 1/3
Salazar-Cavazos 3/18
Okuonghae 2/20
Armistead 1/12
Fujita 1/18
Oliveira 0/20
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of seeds with reduced e_{pred} ≤ full e_{pred}
c
Real data: information inherited from θ_0
θ_0 alone / reduced
Nested reduced / reduced
lower in 20, higher in 2 of 29; median 0.90
lower in 3, higher in 21 of 29; median 1.35
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}
Test χ^2/n relative to the reduced model anchored at θ_0 (values beyond the axis at its edge)